EASY

ALGEBRA

Systems of two linear equations in two variables

02

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ALGEBRA

$$4x - 3y = 5$$

$$x = 8$$

What is the solution x, y to the given system of equations?

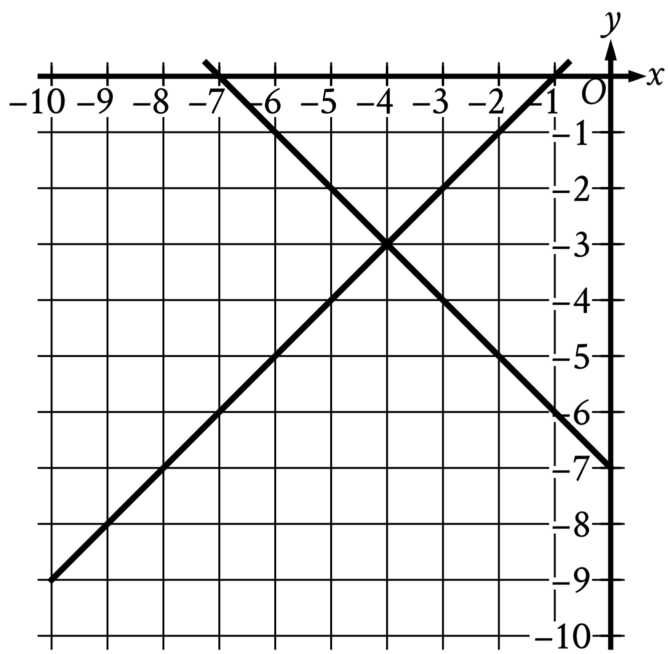
A. $8, 9$

B. $8, -24$

C. $8, -9$

D. $8, 24$

Q2 ALGEBRA



- For the first line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following points:
    - (negative 7 comma 0)
    - (negative 4 comma negative 3)
    - (0 comma negative 7)
- For the second line in the system:
  - The line slants gradually up from left to right.
  - The line passes through the following points:
    - (negative 5 comma negative 4)
    - (negative 4 comma negative 3)
    - (negative 3 comma negative 2)

The graph of a system of linear equations is shown. What is the solution x, y to the system?

A. 0, -7

B. $0, -3$

C. $-4, -3$

D. $-4, 0$

Q3 ALGEBRA

$$5x = 15$$

$$-4x + y = -2$$

The solution to the given system of equations is x, y. What is the value of $x + y$?

A. -17

B. -13

C. 13

D. 17

Q4 ALGEBRA

A discount airline sells a certain number of tickets, x, for a flight for $90 each. It sells the number of remaining tickets, y, for $250 each. For a particular flight, the airline sold 120 tickets and collected a total of $27,600 from the sale of those tickets. Which system of equations represents this relationship between x and y ?

A. $\begin{cases} x+y=120 \\ 90x+250y=27{,}600 \end{cases}$

B. $\begin{cases} x+y=120 \\ 90x+250y=120(27{,}600) \end{cases}$

C. $\begin{cases} x+y=27{,}600 \\ 90x+250y=120(27{,}600) \end{cases}$

D. $\begin{cases} 90x=250y \\ 120x+120y=27{,}600 \end{cases}$

Q5 ALGEBRA

$$y = 2x + 3$$
$$x = 1$$

What is the solution (x, y) to the given system of equations?

A. $(1, 2)$

B. $(1, 5)$

C. $(2, 3)$

D. $(2, 7)$

Q6 ALGEBRA

$$s + 7r = 27$$

$$r = 3$$

What is the solution r, s to the given system of equations?

A. $(6, 3)$

B. $(3, 6)$

C. $(3, 27)$

D. $(27, 3)$

Q7 ALGEBRA

$$4x = 20$$

$$-3x + y = -7$$

The solution to the given system of equations is x, y. What is the value of $x + y$?

A. -27

B. -13

C. 13

D. 27

Q8

GRID-IN ALGEBRA

$$x = 8$$

$$x + 3y = 26$$

The solution to the given system of equations is x, y. What is the value of y?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9 ALGEBRA

$$x = 10$$

$$y = x + 21$$

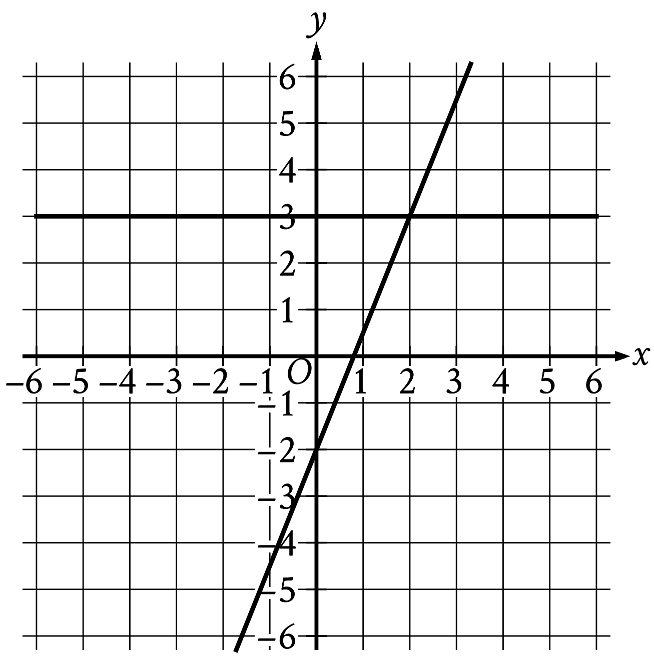
The solution to the given system of equations is x, y. What is the value of y?

A. 2.1

B. 10

C. 21

D. 31



- For the first line in the system:
  - The line is horizontal.
  - The line passes through the following points:
    - (negative 4 comma 3)
    - (0 comma 3)
    - (2 comma 3)
- For the second line in the system:
  - The line slants sharply up from left to right.
  - The line passes through the following points:
    - (negative 1 comma negative 4.5)
    - (0 comma negative 2)
    - (2 comma 3)

The graph of a system of linear equations is shown. What is the solution x, y to the system?

A. $0, 3$

B. $1, 3$

C. $2, 3$

D. $3, 3$